

SCHEDULE 2.

CONSTRUCTION STANDARDS FOR SMALL CRAFT.

For small craft to be registered under this Act the craft must comply with the following standards:

PART I. - SCOPE, DEFINITIONS AND GENERAL CONSTRUCTION REQUIREMENTS.

1. SCOPE.

This Schedule sets out requirements for maximum load, persons and power capacities, and for reserve buoyancy, stability, fire protection and other safety aspects for small craft less than 10 metres in overall length.

2. DEFINITIONS.

For the purpose of this Schedule, the following definitions apply -

“Act” means the ***Small Craft Act 2011***;

“ancillary equipment” means portable or secured items of the boat’s outfitting at a given point in time, such as ground tackle and safety equipment;

“basic flotation” means a flotation system that will keep a Craft carrying its maximum load from sinking when swamped, assuming the occupants of the Craft have left it and are in the water clinging to it;

Note: With basic flotation the swamped Craft may float at any attitude.

“buoyancy” means the force that causes a craft to float, expressed in newtons;

“commercially constructed” a craft constructed by a constructor who builds more than five (5) craft per year;

“craft” means a Small Craft as defined in Section 2 of the Act;

“deck” means the area of a craft that may be walked upon;

“dinghy” means the most common types of open craft operated in Papua New Guinea;

“drain” means a pipe or tube from a deck or cockpit to drain water overboard;

“flotation material” means material with a density less than water, used to provide buoyancy when the craft is swamped;

“freeboard” means the minimum distance between the waterline and the top of the hull

midship at the maximum load capacity but in no case less than 300mm;

“gear” means personal equipment including clothing, provisions and water;

Note: An allowance is made for personal gear (see clause 9).

“length (L)” means the overall length of the boat’s hull;

“level flotation” means a flotation system that will keep a craft carrying its maximum load from sinking when swamped, assuming the occupants remain within the craft and supported by the flotation system. Level flotation implies that the swamped craft will float level and not capsize in calm water, but level flotation does not imply a self-righting capacity;

“maximum load capacity (outboard installations)” means the maximum mass, including motor, accessories and fuel, ancillary equipment, persons and gear that the craft is designed to carry, expressed in kilograms;

“maximum persons capacity” means the maximum load of persons, expressed as the number of average size adults, means that the craft is designed to carry;
 “maximum power capacity” means the power rating of the largest outboard motor suitable for use with the boat. The power rating is expressed in horsepower;
 “mid-length (amidships)” the mid-point of the craft’s length;
 “NMSA” means the National Maritime Safety Authority of Papua New Guinea and its successors;
 “open craft” means a craft not protected from the entry of water by means of a complete weather deck or a partial weather deck and a weather tight or watertight cabin, such that the deck, cockpit or well bottom is open to the bilge and cannot drain overboard;
 “outboard motor” means a self-contained propulsion unit, usually mounted over the stern of a craft;
 “power craft” means a mechanically powered craft;
 “Registrar” means the person appointed as the Provincial Small Craft Registrar as defined in Section 6 of the Act; “reserve buoyancy” means the force that causes a craft to float when swamped, expressed in newtons;
 “sheer” means the fore-and-aft curve of a craft’s deck;
 “static float plane” means a plane parallel to the reference plane passing through the lowest point of the weather deck;
 “stability” means the ability of the craft to return to its normal attitude;
 “watertight” means constructed to provide effective protection against water seepage when closed and exposed to continuous driving rain or waves;
 “weather deck” means the uppermost continuous deck exposed to the weather or, if the craft is not fully decked, the uppermost partial deck exposed to the weather;
 “weather tight” means constructed to provide effective protection against water seepage when closed and exposed to ordinary rain or spray.

3. GENERAL CONSTRUCTION REQUIREMENT.

(1) All craft submitted for registration under the Act shall comply with the relevant provisions of the schedule before registration is granted -

- (a) for a craft constructed before the Act came into force: the craft, in the opinion of the Registrar, is of suitable construction and in suitable condition to be safely operated for its intended purpose at sea; or
- (b) for new craft commercially constructed prior to 1 October, 2016: in the opinion of a Registrar or the NMSA, is of suitable construction and design to be safely operated for its intended purpose at sea; or
- (c) all Craft commercially constructed after 1 October, 2016:
 - (i) certification by the NMSA that the craft complies with the requirements of this schedule; or
 - (ii) certification of that the craft or class of craft complies with the approved construction standards of a foreign maritime authority and approved by the NMSA; or
 - (iii) certification by an NMSA recognised marine surveyor or marine architect that the craft is of suitable construction and design to be safely operated for its intended purpose at sea and complies with the requirements of this schedule.

(2) The NMSA may issue a certificate of compliance with construction standards for a class or model of a commercially built craft and such certification is to be considered as compliance with this standard by all provincial small craft board registrars.

PART II. - MAXIMUM CAPACITIES AND BOUANCY.

Division 1. - Maximum Load Capacity for Crafts.

4. GENERAL.

The maximum load capacity shall be -

- (a) for a dinghy, using the maximum relevant load capacity for the length of the dinghy contained in Table 1 factoring in the outboard motor size and weight of crew and or passengers; or
- (b) determined in accordance with Appendix A of Australian Standard AS1799.1
Method for Calculating Maximum Load Capacity and Clause 2.1.

5. OUTBOARD INSTALLATIONS.

The maximum load capacity shall be calculated in kilograms, and shall be determined by calculating or measuring the cubic capacity below the static float plane, converting this volume to the mass of water it would displace, subtracting the mass of the craft, including the filled mass of any installed fuel tanks, and allowing 1 kg of load capacity for each 5 kg of remaining displacement.

6. INBOARD INSTALLATIONS.

The maximum load capacity shall be calculated in kilograms, and shall be determined by calculating or measuring the cubic capacity of the craft below the static float plane, converting this volume to the mass of water it would displace, subtracting the mass of the craft, excluding the mass of the engine, fuel tank and fuel, allowing 1 kg of craft load capacity for each 5 kg of remaining displacement, and finally subtracting the mass of the engine, fuel tank and fuel to obtain the maximum load capacity.

Division 2. - Maximum Persons Capacity.

7. GENERAL.

The maximum person capacity provides a quick guide to the number of adults that can be safely carried when operating. It is derived using an average value for the mass of a passenger and the allowance for the mass of ancillary equipment and personal gear on board. It should therefore be treated with some caution and where doubt exists, the maximum load capacity should be used as the indicator of the boat's true capacity. The maximum persons capacity for protected shall be determined as follows:

- (a) for a dinghy, using the maximum relevant capacity number for the length of the dinghy contained in Table 1 factoring in the outboard motor size; or
- (b) by estimating the persons capacity and modifying this value, as necessary, to comply with the appropriate stability requirements of Clause 31.

8. OUTBOARD INSTALLATIONS.

The maximum person capacity (c) for craft shall be stated as the number of average adults and shall be determined by the following method:

- (a) take the maximum load capacity and subtract the assumed mass of the largest outboard motor (including controls) for which the craft is rated including the mass of batteries, obtained from Table 2. For craft without installed fuel tanks, also subtract the mass of the portable fuel tank and fuel; and
- (b) divide the result by 75 kg, and take the lower whole number, thus obtaining the maximum number of average size adults.

9. INBOARD INSTALLATIONS.

The maximum persons capacity (c) shall be determined by taking the maximum load capacity, dividing by 75 kg and taking the lower whole number, thus obtaining the maximum number of average size adults.

Note: An allowance of 70 kg is made for the body mass of each adult with an additional allowance of 5 kg per person for ancillary equipment and personal gear.

Division 3. - Swamped Flotation.

10. MINIMUM SWAMPED FLOTATION.

All craft, except those over 6 meters in length that are fully enclosed and watertight shall be provided at least with basic flotation when swamped in the form of flotation material (closed-cell plastics or equivalent) or air chambers. Basic flotation shall be demonstrated by a practical test to the satisfaction of the NMSA. For basic flotation, at least some portion of the hull shall remain above water when the craft is swamped in a condition of maximum load capacity (as nominated on the Builders Plate), when fitted with the largest motor for which the craft is rated (for outboard engine craft), and with all ancillary equipment and gear in their normal position.

Note: The risk of a craft less than 6 metres in length being swamped without warning is considered to be sufficiently high that reliance cannot be placed on closing up watertight openings to provide reserve buoyancy.

11. USE OF FLOTATION MATERIALS.

Flotation materials are to be located in a position where it is unlikely that they will come into contact with petroleum products or other chemicals used in the construction or operation of the boat, such as within an elevated thwart on an aluminum dinghy. Where flotation materials are used, an allowance of 10% above the minimum determined for new materials shall be added to cater for the effects of shrinkage and loss of buoyancy over time.

12. RESISTANT FLOTATION MATERIALS.

Resistant flotation materials shall not lose more than 10% of their buoyancy when tested for water absorption, resistance to heat and resistance to fuels in accordance with

IMO MSC 81(70). Polyethylene, polypropylene and polyurethane buoyancy foams complying with commercial vessel requirements of the NSCV shall be deemed to satisfy the requirements.

Note: Polystyrene foam does not meet the requirements.

13. LEVEL FLOTATION.

In addition to meeting the requirements for minimum swamped flotation, all craft shall be provided with level flotation. Level flotation shall be demonstrated by a practical test to the satisfaction of the NMSA. The NMSA may certify a craft or class of craft as meeting this requirement after such a practical test.

14. AIR COMPARTMENTS.

(1) Wherever practicable, integral void air compartments should be avoided as a means of providing reserve buoyancy. Where such compartments are used, the construction shall be equivalent to that of the surrounding hull structure. Stress raisers shall be avoided, and all compartment-to-hull seams shall be positioned to allow inspection. Where air chambers or compartments are used to provide reserve buoyancy, the requirements shall be met excluding the two largest compartments or air chambers. Each air compartment used to provide reserve buoyancy shall be –

- (a) provided with a drain plug at its lowest point; and
- (b) permanently marked with the following words or their equivalent:

Caution: This air compartment is essential to the flotation of the boat. Do not puncture or attach fittings.

(2) These requirements apply to all of the air compartments used to provide reserve buoyancy, including the two largest air compartments.

15. MAXIMUM POWER CAPACITY.

(1) Determination of maximum power capacity (outboard installations). The maximum power capacity of craft shall be stated in horsepower and shall be determined by the following method (see Table 3 at rear of schedule) -

- (a) calculate the factor for use in Table 3 by multiplying the overall craft length in metres by the transom width in metres. The transom width shall not exceed the widest measurement of the transom in that part which is below the point of ingress of water. If spray rails act as chines or part of the planning surface, they may be included in the transom width, but otherwise fins and flare shall be excluded. Where a craft has a rounded stern, for the transom width substitute the maximum width below the static float line measured at a point one quarter of the craft length forward of the stern; and
- (b) use the factor to determine the corresponding maximum power capacity direct from Table 3, or, if the factor is over 5, calculate the maximum power capacity by multiplying the factor and subtracting as shown in the table. If the factor is over 5, and the calculated power capacity is not a multiple of 5, the calculated power capacity may be raised to a multiple of

5 to accommodate power ranges of stock engines. For flat-bottomed hard chine craft, the maximum power capacity shall be reduced by one increment (of power capacity in Table 3) for factors below 5

16. MARKING.

Craft capacities -

- (a) Builders Plate - Craft shall be fitted with a Builders Plate approved by the NMSA; and
- (b) record the maximum power, load and persons capacity for the craft on the plate; and
- (c) the capacities shown must not be greater than that determined in accordance with this Standard;
- (d) Every craft commercially constructed must have a constructor's serial number and construction completion date for each craft recorded on the Builders Plate.

Note: The constructor may nominate a lower maximum load and persons capacity than the value calculated. This reduction may be desirable for reasons, such as limitations on seating and accommodation or a decision that it would be prudent not to approach the boat's limits.

PART III. - HULL DESIGN.

Division 1. - General Arrangement.

17. GENERAL ARRANGEMENT.

The design of a craft should take into account all aspects of construction, power, accommodation, access and egress, and the stowage of equipment in the safest and most effective manner. Reference should be made to recognised national and international standards when considering structural and engineering design. Where craft are fitted with inboard engine installations, reference should be made to recognised national and international standards for ventilation requirements.

Division 2 - Hull Drainage.

18. BAILING DEVICES.

- (1) Each craft shall be fitted with an efficient bilge-pumping, except in the following cases -
 - (a) an open craft of length less than 10 metres where there is no subdivision of the bottom (e.g. by means of floors), in which case buckets or bailers may be used in lieu of a bilge pumping system; and
 - (b) a fully self-draining craft that may be capsized through 360 about a horizontal axis without allowing the entry of any water below its weather deck and in which all spaces below the weather deck are used solely as buoyancy compartments.

(2) Craft not fitted with bilge pumping arrangements shall be fitted with adequate drain plugs to permit water to be drained from the hull. In that case, suitably placed, accessible limber holes of adequate size should be provided to enable the hull to be efficiently drained without excessive tilting. Venturi or other similar types of automatic bailing devices shall be of such a type as to prevent flooding by syphon effect or when the craft is stopped.

19. COMPARTMENTS.

Craft with watertight or weather tight compartments shall be provided with drainage from each compartment. The arrangement shall be such that water cannot penetrate from one watertight or weather tight compartment to another through the bilges when the drain plugs are in place.

20. DRAIN PLUGS.

Drain plugs may be either threaded or have an equivalent mechanical means of sealing, and shall comply with the following requirements:

- (a) the drain plug fitting shall be designed and installed so that with the drain plug correctly fitted the watertight integrity of the hull is not compromised; and
 - (b) drain plugs shall be captive, readily accessible for inspection, and easily operable by hand; and
 - (c) plugs and barrels shall be manufactured of corrosion-resistant materials;
- and
- (d) the means provided to lock the plug in position shall have sufficient friction to prevent accidental removal; and
 - (e) friction-fit plugs shall not be used; and
 - (f) elastomeric sealing members should be compounded to resist attack from salt water, petroleum products, sunlight, ozone, marine growth or reaction with metal components. The sealing member should also resist compression set and maintain sufficient resilience to return quickly to the relaxed form for easy removal from the hull. It should not adhere to the drain tube.

Division 2. - Decks.

21. GENERAL.

Weather decks shall be watertight. Weather decks should be provided with sheer or camber. Excessive sheer or camber should be avoided on frequently used decks.

22. FOOTING.

Weather deck areas intended to be frequently walked upon shall have a slip-resistant surface, and have at their outboard edges a toe rail or other suitable means to help prevent loss of footing.

Division 3. - Transom for outboard or stern drive installations.

23. TRANSOM.

The transom shall be capable of absorbing and transmitting to the hull of the craft all of the loadings imposed on it by the maximum power motor recommended for the craft as well as the loadings imposed while being trailed.

Note: Support for the motor may be provided on the trailer.

Division 4. - Motor Well.

24. MOTOR WELL.

The motor well, where provided, shall raise the static float plane above the transom cutout. Steering and motor accessory ports shall be located as high as possible with an aggregate area as small as possible but not exceeding 100 cm², and shall be capable of being protected against excessive water entry. The well shall be watertight to the hull interior and have drain holes through the transom of sufficient size to allow water to drain rapidly. The motor well shall be dimensionally suitable for all motors it is intended to carry, and shall comply with the requirements of the motor manufacturer in this regard.

Division 5. - Craft Hardware and Fittings.

25. GENERAL.

- (1) Craft hardware and fittings shall comply with the following requirements -
 - (a) be of sufficient strength to withstand the maximum loads likely to be applied in normal and emergency service; and
 - (b) be of a size and design to permit easy use, particularly with regard to the attachment of lines; and
 - (c) be resistant to deterioration by corrosion or weather; and
 - (d) be free of sharp edges or dangerous features that could cause injury.
- (2) Consideration should be given to the possibility of injury in a collision or in severe conditions.
- (3) All load-bearing fittings shall be welded or through-fastened with bolts (or other positive through fastening methods) and secured so that they cannot become damaged or work loose in service. All fittings shall be adequately backed with reinforcements so that the applied loads will be distributed to the adjacent structures.

26. BOW EYE.

Each craft shall be fitted with a bow eye or other fitting suitable for use in towing by other craft, or for attachment to the bow post of a trailer. The bow eye shall be able to withstand a direct tensile load equivalent to twice the weight of the fully loaded craft. The bow eye shall be located above the waterline and shall be through-fastened with bolts or welded to the stem. Where such eyes are not accessible from within the craft in adverse weather or because of raked stems or for other reasons, an additional towing point, accessible from within the craft, shall be provided.

27. DISSIMILAR METALS.

Except where intended for cathodic protection, dissimilar metals shall not be used in contact with each other in the craft unless suitable precautions are taken to minimize corrosion.

Note: Guidance on the selection of suitable alloys is given in AS 1799.3.

PART IV. - VISABILITY.

28. GENERAL.

Each craft shall have a sufficient area of hull coloured in an NMSA approved marine orange colour to enable it to be sighted from a distance of not less than three kilometres.

PART V. - STABILITY.

29. GENERAL.

Craft shall conform to the stability requirements relevant for craft contained in Section 5 - Stability, of Australian Standard 1799.1

PART VI. - FIRE PROTECTION.

Division 1. - Portable fire extinguishers.

30. SELECTION.

The type of fire extinguishers to be carried on a craft shall be appropriate to the type of fires likely to be encountered.

31. GENERAL.

All boats except open craft that are powered by an outboard motor should carry at least one extinguisher.

33. LOCATION AND MOUNTING.

Portable fire extinguishers shall be located so they are easily identifiable and readily accessible. Portable fire extinguishers shall be mounted in such a way that they will not come adrift during operation of the boat, and can be readily demounted for use when required.

34. INBOARD ENGINE COMPARTMENTS.

On inboard boats where an automatic fire extinguishing system is not installed, provision shall be made so that a portable fire extinguisher can be discharged into the engine space while the engine cover remains closed.

35. EXEMPTION

The NMSA may by written notice, exempt a class or type of small craft or individual small craft from any or all provisions and requirements of this Schedule.

TABLE 1.

STANDARD CALCULATIONS FOR MAXIMUM PASSENGER AND LOAD CAPACITY OF PNG DINGHIES.

Dinghy overall length (metres).	Dinghy overall length (feet).	Maximum Outboard Motor Size.	Weight of Motor (kg).	Maximum Passenger loading.	Maximum permissible weight (kg) (including motor).
5.5-5.99	19	30hp	53	6	790
7.0-7.49	23	40hp	76	8	970
7.5-7.99	24	50hp	100	12	1500

TABLE 2.

MASSES OF PETROL OUTBOARD MOTORS AND RELATED EQUIPMENT FOR VARIOUS MAXIMUM POWER CAPACITIES.

1		2	3	4	5	6	7
Maximum power capacity				Mass, kg			
HP	kW	Motor and controls	Battery	Portable fuel tank and fuel	Total Mass	Submerged mass	Swamped mass
0-2.0	0-1.5	15	0	0	15	10	12
2.1-3.9	1.6-2.9	18	0	0	18	12	15
4.0-7.0	3.0-5.2	41	0	11	52	30	35
7.1-15.0	5.3-11.2	60	10	22	92	55	50
15.1-25.0	11.3-18.7	108	20	22	150	90	90
25.1-45.0	18.8-33.6	125	20	45	190	115	125
45.1-60.0	33.7-44.8	165	20	45	230	140	140
60.1-75.0	44.9-56.0	190	20	45	255	155	160
75.1-100.0	56.1-74.6	210	20	45	275	165	175